INCH-POUND

MIL-G-50678C (AR)
AMENDMENT 1
24 June 1991

MILITARY SPECIFICATION

GATE, MULTIPLE, CIRCUITS: 9342754

This amendment forms a part of Military Specification MIL-G-50678C (AR), dated 30 April 1990, and is approved for use by the U.S. Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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2.1.1, Delete: "MIL-S-45743 - Soldering, Manual Type High Reliability Electrical and Electronic Equipment" and "MIL-S-46844 - Solder Bath Soldering of Printed Wiring Assemblies".

ADD: "MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies".

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3.2, Delete: "e. Soldering electrical connections."

PAGE 6

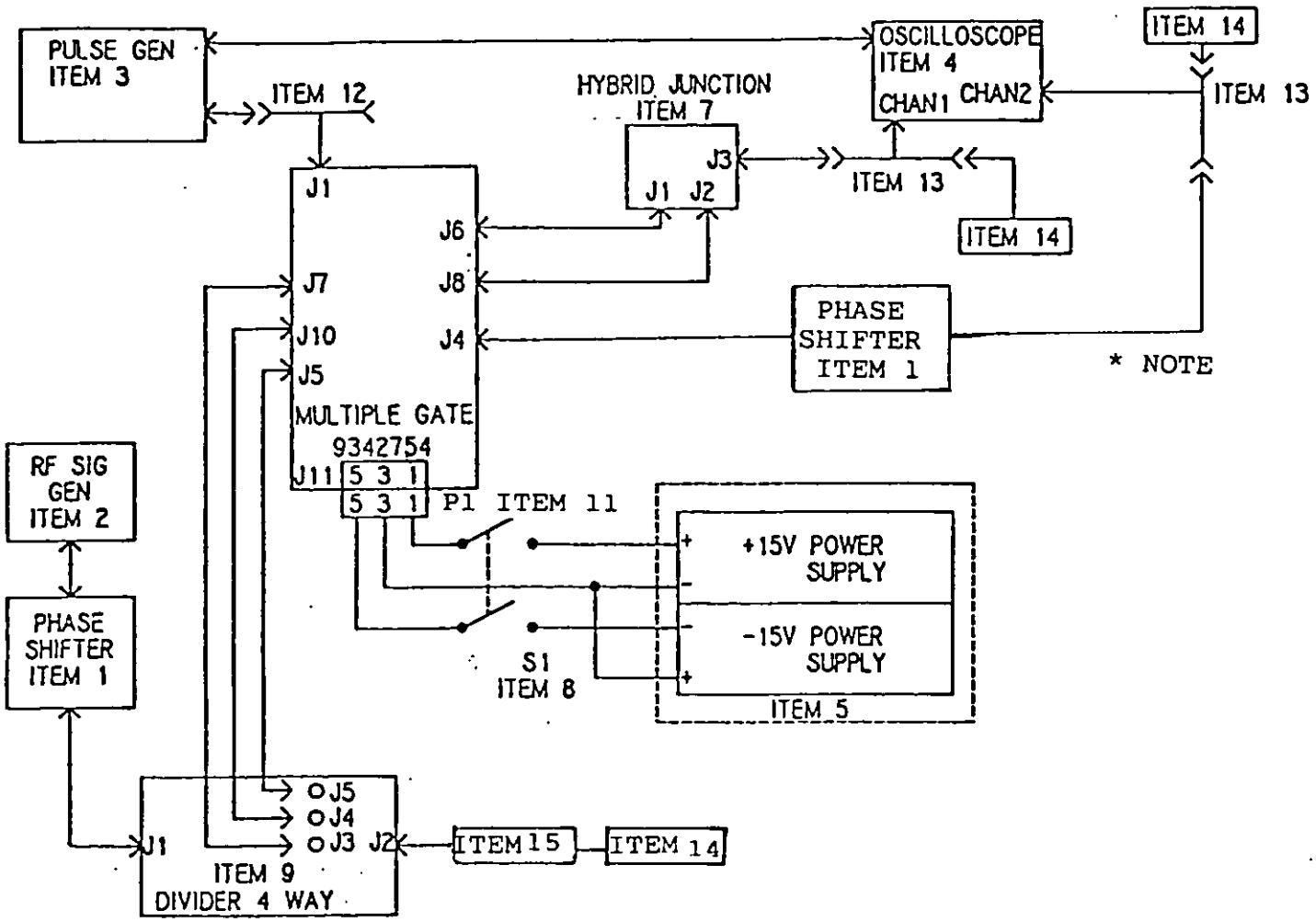
Add new paragraph 3.5.2.3.4 as follows:

"3.5.2.3.4 Gate Noise. With the multiple gate connected with the test setup as shown in Figure 5, a 60± 1% MHz signal and a monitored 2.8 V p-p signal per gate applied to J5, J7 and J10 of the unit under test, the amplitude of the algebraic sum of the Early Gate and Sum Gate signals during the Early Gate period shall be less than 1.0 V p-p. The amplitude of the algebraic sum of the Late Gate and inverted Sum Gate signals during the Late Gate period shall be less than 1.0 V p-p."

Add Figure 5. enclosed (page 6A).

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* NOTE:

Adjust Channel 2 CAble length as necessary to place Channel 1/
/Channel 2 Wave form relationship into phase shifter adjustment range.

FIGURE 5. Multiple gate circuit test setup.

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3.7, Delete in its entirety and substitute the following:

"3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task)F of paragraph 1.2 of MIL-STD-2000."

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4.6.2.3, Table III, Item 1, under quantity. Delete "1. Phase shifter Model PS-3-60, Merrimac or equal." Add "1. Phase Shifter Model PS-3-60, Merrimac or equal (2)."

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Add new paragraph 4.7.3.4.4 as follows:

"4.7.3.4.4 Gate noise. Connect the unit under test to the test setup per Figure 5. Adjust channel 2 Phase Shifter so that the Early Gate on channel 1 is 180° out of phase with the Sum Gate on channel 2. Adjust variable gain on channel 2 for equal amplitude waveforms on channels 1 and 2. Place oscilloscope in channel 1 and 2 algebraic add function and readjust channel 2 phase shifter for Early Gate null. Resultant waveforms in the Early Gate shall be less than 1.0 V p-p. Place oscilloscope channel 2 in invert mode. Readjust amplitude and phase for Late Gate null. Resultant waveforms in the Late Gate shall be less than 1.0 V p-p."

Add paragraph 4.8 as follows:

"4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7."

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